HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed Python data pipelines for ingesting and processing large volumes of audio data, optimizing for throughput and cost for AI model training datasets.
- Built and maintained production-grade data processing systems on AWS, ensuring data quality and scalability for petabyte-scale datasets.
- Collaborated with AI Scientists to refine training datasets and support the development of next-generation AI models, translating research needs into engineering tasks.
- Designed and implemented automated data acquisition workflows from new audio sources, integrating them into existing data infrastructure.
- Containerized data processing services using Docker and deployed them to an AWS environment, enabling more efficient resource management and faster iteration cycles.
- Managed engineering projects from initial problem discovery through implementation and continuous improvement, ensuring timely delivery and adherence to quality standards.
- Analyzed and resolved production incidents related to data pipelines, performing root cause analysis and implementing preventative measures to enhance system reliability.

SKILLS

Programming Languages – Python, Bash

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch), GCP

Data Engineering – Data Pipelines, Large-Scale Data Processing, ETL, Data Acquisition, Web Crawlers

ML / AI – AI Training Data, Dataset Quality, Audio Processing, Machine Learning Fundamentals

Infrastructure – Docker, Terraform, Linux

Development Tools – Git, Jenkins

PROJECTS

Audio Data Ingestion Pipeline for ML Training

- Engineered a Python-based pipeline to discover and ingest diverse audio data sources, handling approximately 5 TB of raw data daily for ML model training.
- Implemented data validation and cleaning steps within the pipeline to ensure high dataset quality, reducing downstream processing errors by an estimated 15%.
- Deployed the pipeline on AWS Lambda and S3, monitoring performance and cost-efficiency for continuous operation.

Scalable Data Processing System for Speech Analysis

- Developed a distributed data processing system using Python and Bash on Linux to handle large-scale audio feature extraction for AI research.
- Optimized processing logic to achieve a throughput of ~1,500 audio files per hour, supporting the needs of the AI research team.
- Integrated the system with cloud infrastructure (GCP) using Terraform for automated provisioning and management.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)